## 1️⃣ Interpret 5 (Most Relevant)

**Interpret mathematical outcomes in a variety of formats in relation to a situation.**

### Reason

The task involves:

- the original model (a 3D visual representation)
- multiple candidate images (2D representations from different viewpoints)

Students need to:

👉 interpret and compare across different representations

For example:

- infer the back view from the front/side view
- determine the positions of the window and the door after viewpoint transformation

This reflects the ability to:

**interpret and compare multiple representations (views)**

Therefore, this is the **core competency** in this problem.

## 2️⃣ Interpret 9

**Understand the relationship between the context of the problem and the representation of the mathematical solution.** interpret

### Reason

The real-world context is:

👉 a physical garage structure

The mathematical representation is:

👉 geometric drawings from different viewpoints

Students must understand:

- real object ↔ graphical representation
- relationships between different viewpoints

For example:

👉 the "back view" may involve changes in left-right orientation

This reflects:

**understanding the mapping between context and geometric representation**

## 3️⃣ Interpret 1

**Interpret a mathematical result back into the real world context.** interpret

### Reason

After selecting an option (A/B/C/D), students must interpret:

👉 whether the chosen diagram correctly represents the real garage

That is:

- graphical judgment → real-world structure

This reflects:

**interpreting a result back into the real-world context**

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## 4️⃣ Interpret 2

**Identify whether a mathematical result makes sense given the context of a problem.** interpret

### Reason

Students must evaluate:

👉 whether each option is reasonable

For example:

- Is the window on the correct side?
- Should the door be visible from the back?

Options that do not match the real structure must be eliminated.

This reflects:

**evaluating the reasonableness of results in context**

## 5️⃣ Interpret 8 (Less Relevant)

**Recognise the extent and limits of mathematical concepts and solutions.** interpret

### Reason

Students need to recognize:

- limitations caused by viewpoint changes
- incomplete information in the given representation

For example:

👉 the front view does not directly provide full information about the back

This reflects:

**understanding the limitations of representations**

## 1️⃣ Interpret 5（相关性最高）

**Interpret mathematical outcomes in a variety of formats in relation to a situation.**

### 体现原因

题目涉及：

- 原始模型图（3D视觉表示）
- 多个候选图（不同视角的二维表示）

学生需要：

👉 在不同表示之间进行理解和比较

例如：

- 从正面/侧面 → 推断背面
- 判断窗户和门的位置变化

这体现了：

**在多种表示形式（视图）之间进行解释与比较的能力**

因此是本题最核心能力。

## 2️⃣ Interpret 9

**Understand the relationship between the context of the problem and representation of the mathematical solution.** interpret

### 体现原因

现实情境是：

👉 一个真实的车库结构

数学表示是：

👉 不同视角的几何图形

学生需要理解：

- 现实物体 ↔ 图形表示
- 不同视角之间的对应关系

例如：

👉 “背面视角”意味着左右可能发生变化

体现了：

**情境与几何表示之间的映射理解**

## 3️⃣ Interpret 1

**Interpret a mathematical result back into the real world context.** interpret

### 体现原因

学生最终选择某一选项（如 A/B/C/D），需要理解：

👉 该选项是否真实对应原车库结构

即：

- 图形判断结果 → 对应真实结构

体现了：

**从图形判断回到现实模型的解释能力**

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## 4️⃣ Interpret 2

**Identify whether a mathematical result makes sense given the context of a problem.** interpret

### 体现原因

学生需要判断：

👉 某个选项是否合理

例如：

- 窗户是否出现在正确的一侧
- 门是否在背面可见

若不符合现实结构，则排除。

体现了：

**对结果合理性的情境判断能力**

## 5️⃣ Interpret 8（次相关）

**Recognise the extent and limits of mathematical concepts and solutions.** interpret

### 体现原因

学生需要意识到：

- 视角变化带来的限制
- 图形信息的不完全性

例如：

👉 正面图无法完全直接给出背面信息，需要推断

体现：

**对表示局限性的理解**